

CHEM 210

Modern Applications of Chemistry

Duke University

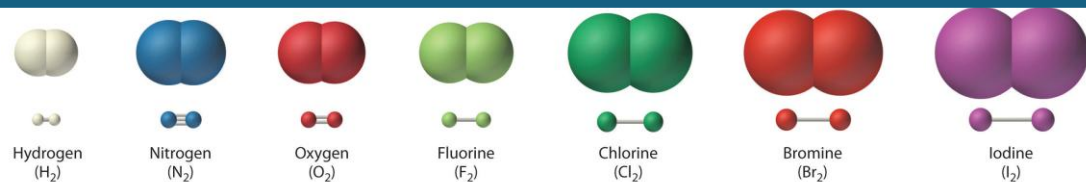
Prof. Charlie Cox

Topics:

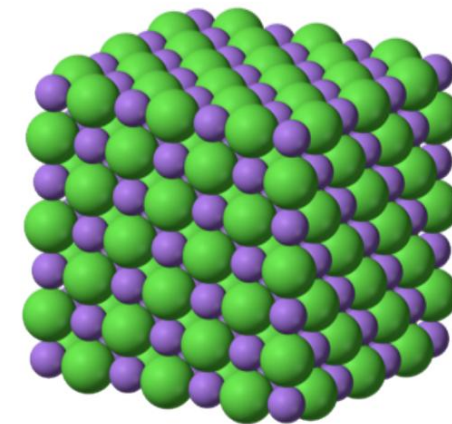
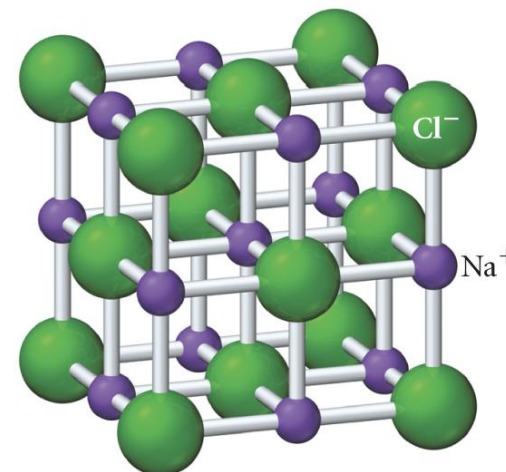
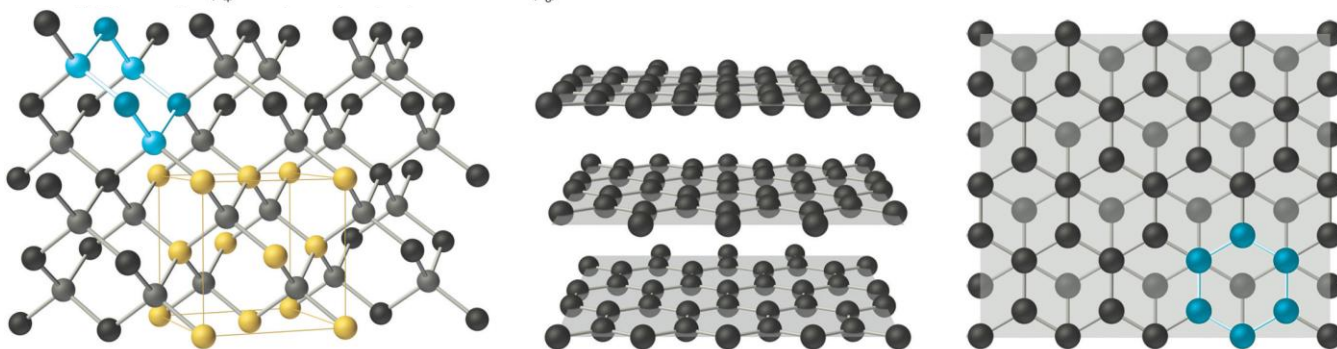
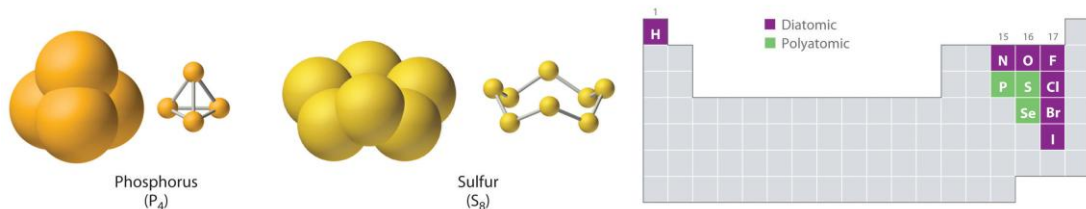
- 1.) Intermolecular Forces
- 2.) Phase Changes
- 3.) Vapor Pressure
- 4.) Energetics of Phase Changes



Types of Chemical Bonds

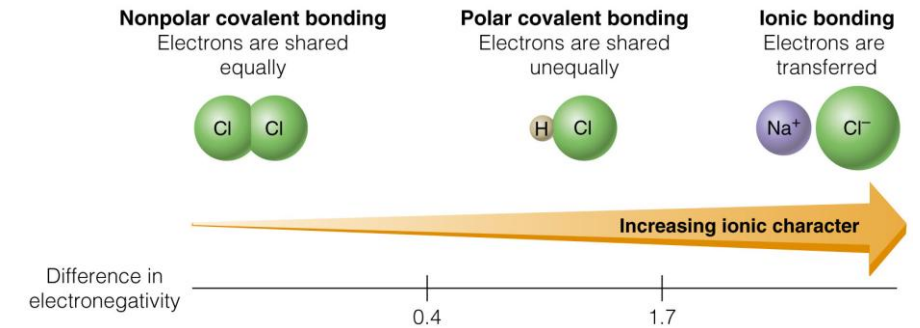
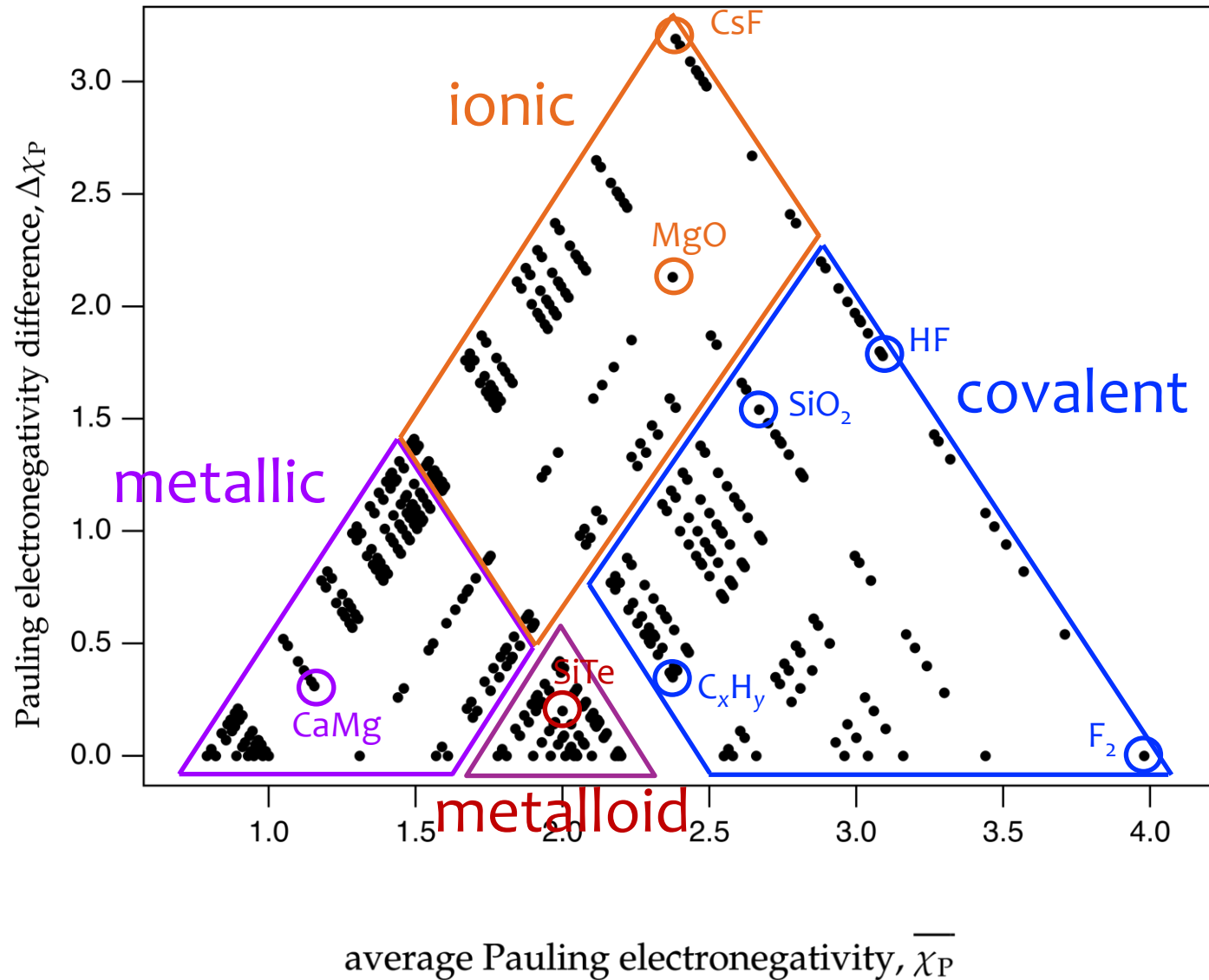


(a) Elements that exist as diatomic molecules

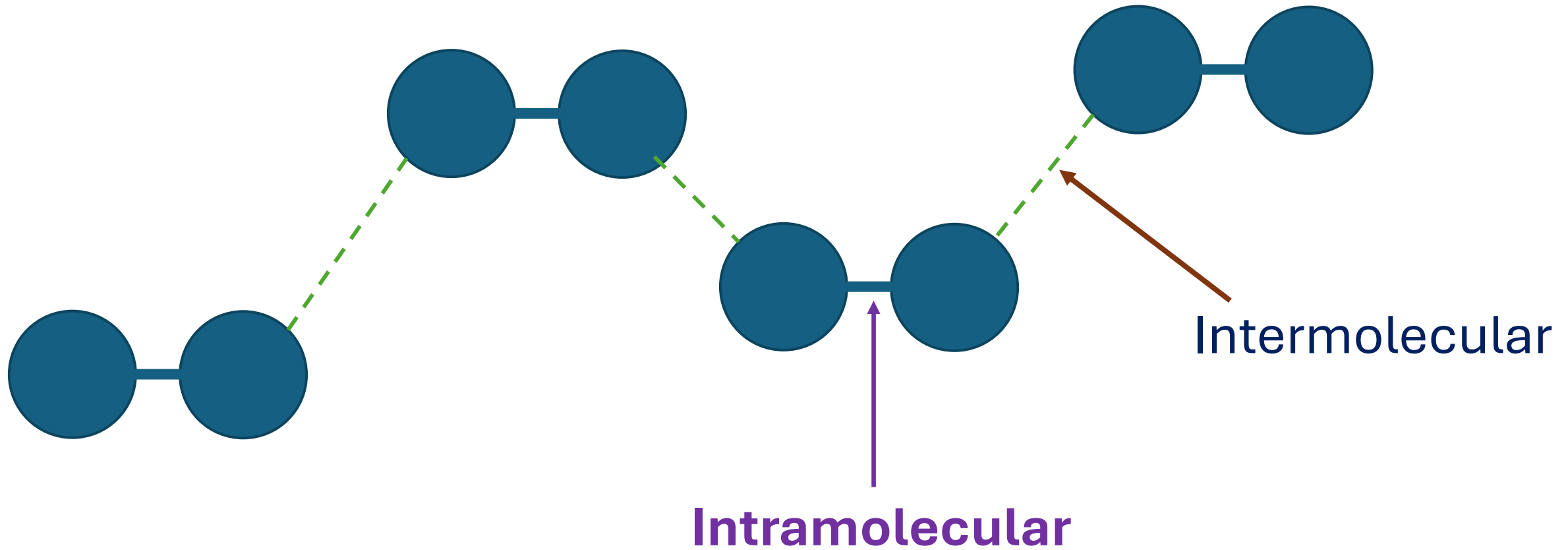


elemental form	H ₂	He	Li(c)	Be(c)	B(c)	C(c)	N ₂	O ₂	F ₂	Ne
melting point	13.81 K	0.95 K	453.65 K	1560 K	2348 K	3823 K	63.15 K	54.36 K	53.53 K	24.56 K
boiling point	20.28 K	4.22 K	1615 K	2744 K	4273 K	4098 K	77.36 K	90.20 K	85.03 K	27.07 K
bp-mp	6.47 K	3.27 K	1161 K	1184 K	1925 K	275 K	14.21 K	35.84 K	31.5	2.51 K
name	hydrogen	helium	lithium	beryllium	boron	carbon	nitrogen	oxygen	fluorine	neon

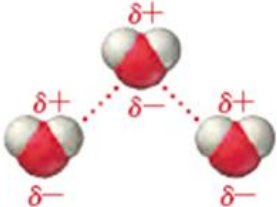
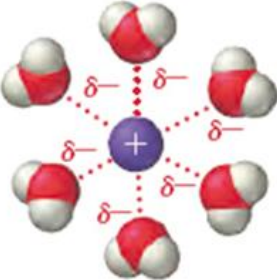
The Bonding Triangle



Intramolecular vs. Intermolecular



Types of Intermolecular Attractions

Type	Present In	Molecular Perspective	Strength
Dispersion*	All molecules and atoms		0.05–20+ kJ/mol
Dipole–dipole	Polar molecules		3–20+ kJ/mol
Hydrogen bonding	Molecules containing H bonded to F, O, or N		10–40 kJ/mol
Ion–dipole	Mixtures of ionic compounds and polar compounds		30–100+ kJ/mol

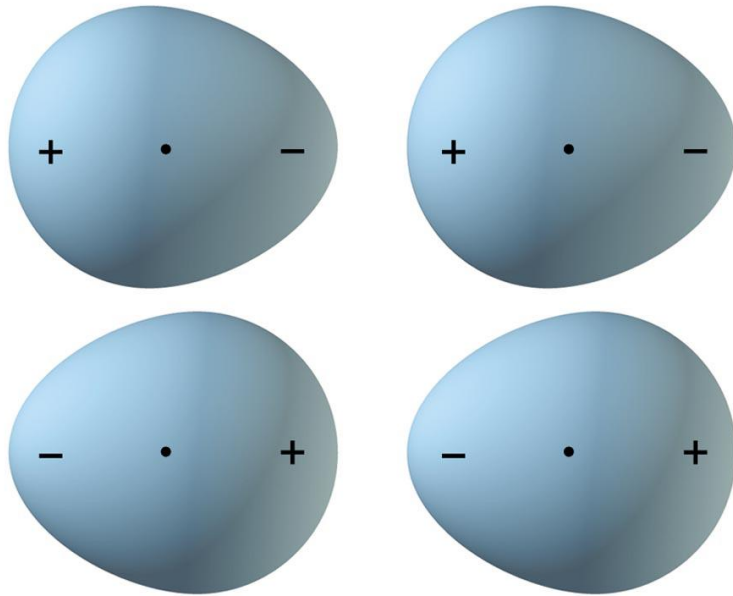
*The dispersion force can become very strong (as strong and even stronger than the others) for molecules of high molar mass.

Are we limited to a single type of intermolecular attraction within a molecule?

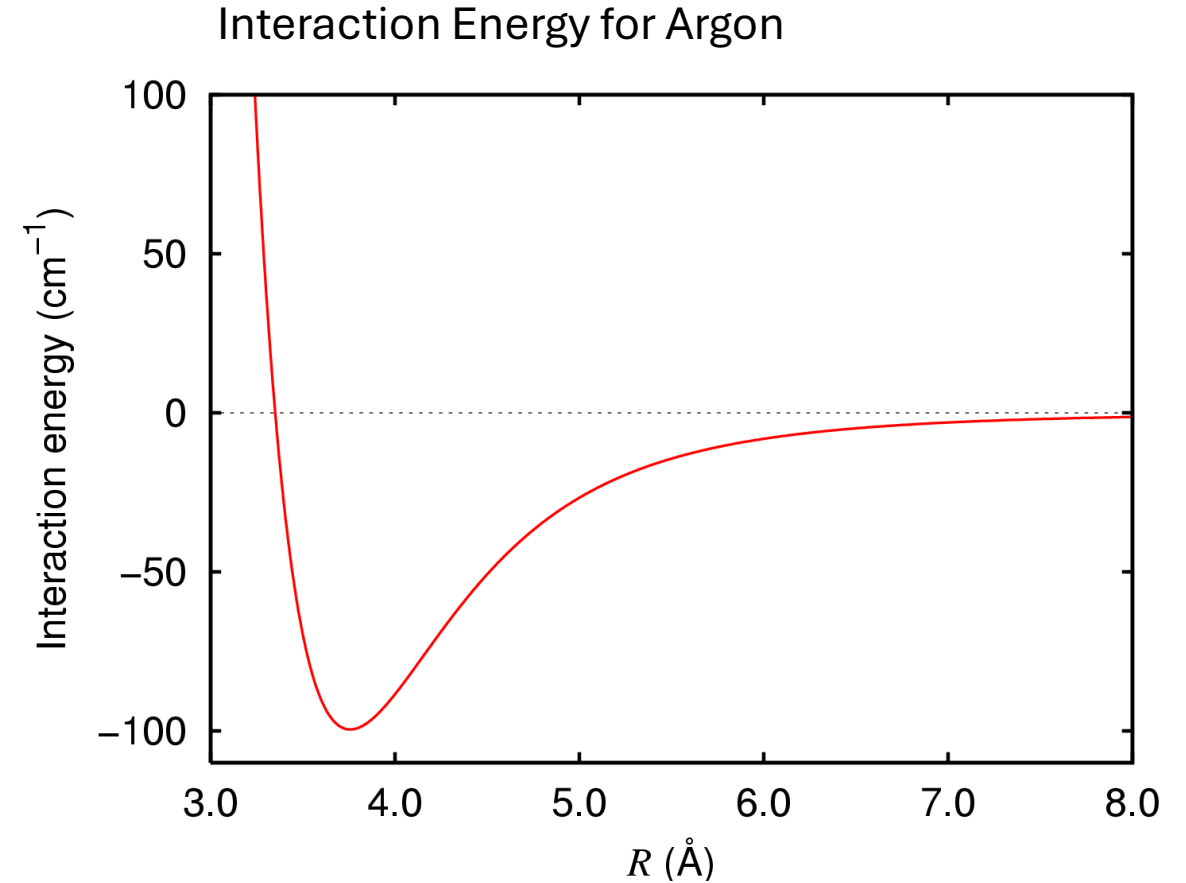
London Dispersion

Electron clouds are polarizable.

What does this mean?



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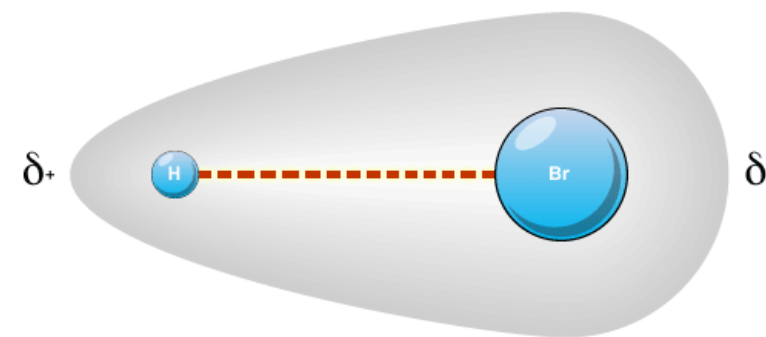
London Dispersion and Mercury

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Dipole-Dipole Interactions

- Recall polar molecules. What is observed?
- How is this similar to London Dispersion?
- How is this different from London Dispersion?
- Which requires more energy to break?



Hydrogen Bonding

When H bonds directly to F, O, or N, the bonding atoms acquire relatively large partial charges, giving rise to strong dipole-dipole attractions between neighboring molecules.



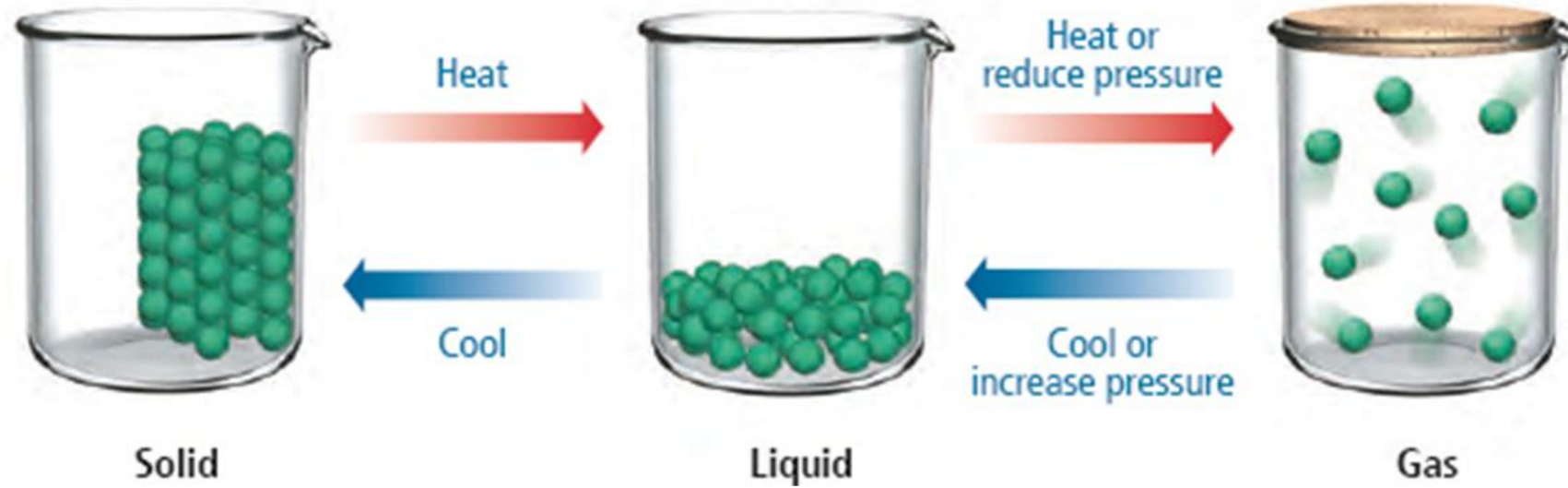
Space-filling model



Electrostatic potential map

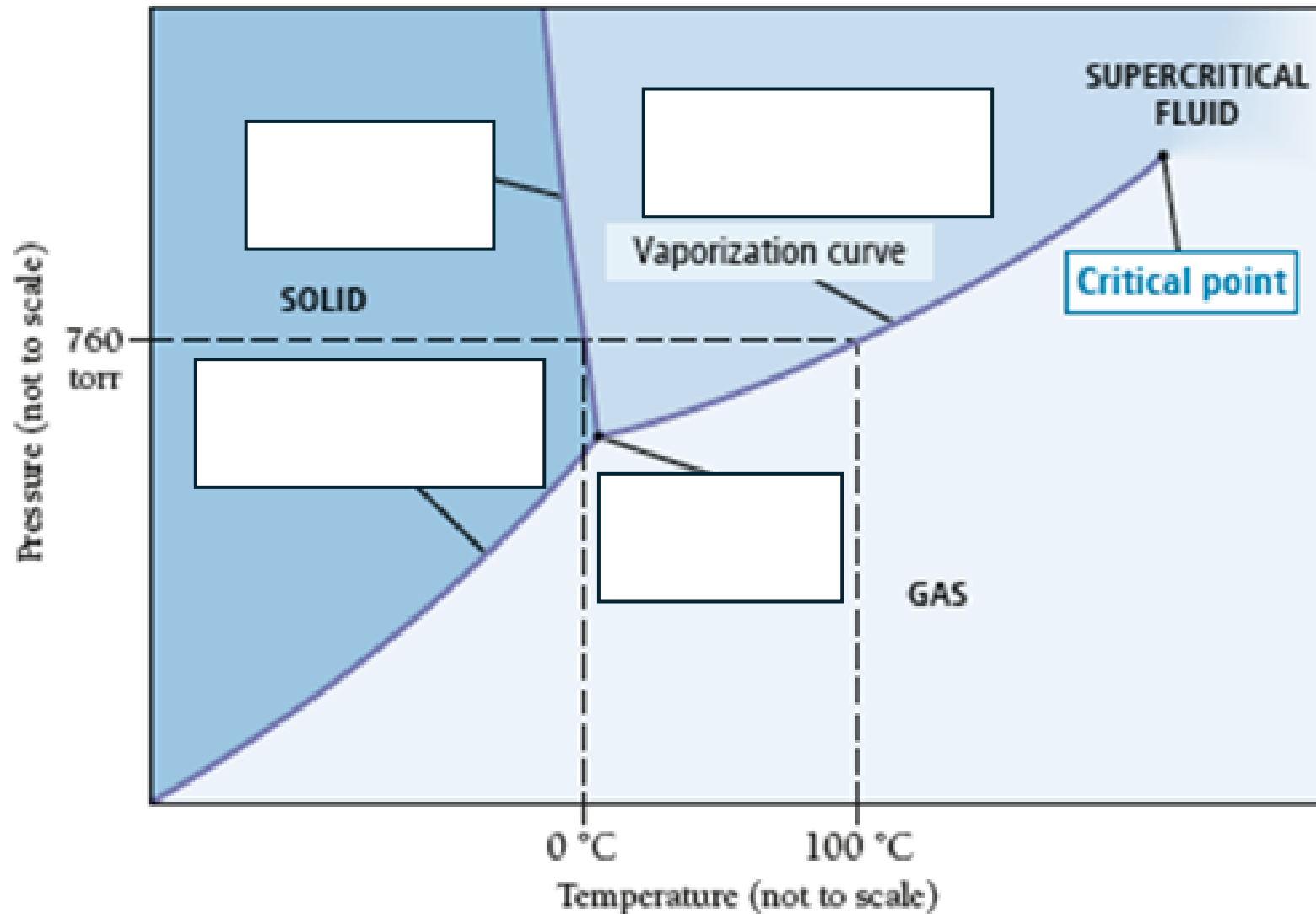
Intermolecular Forces and Phase

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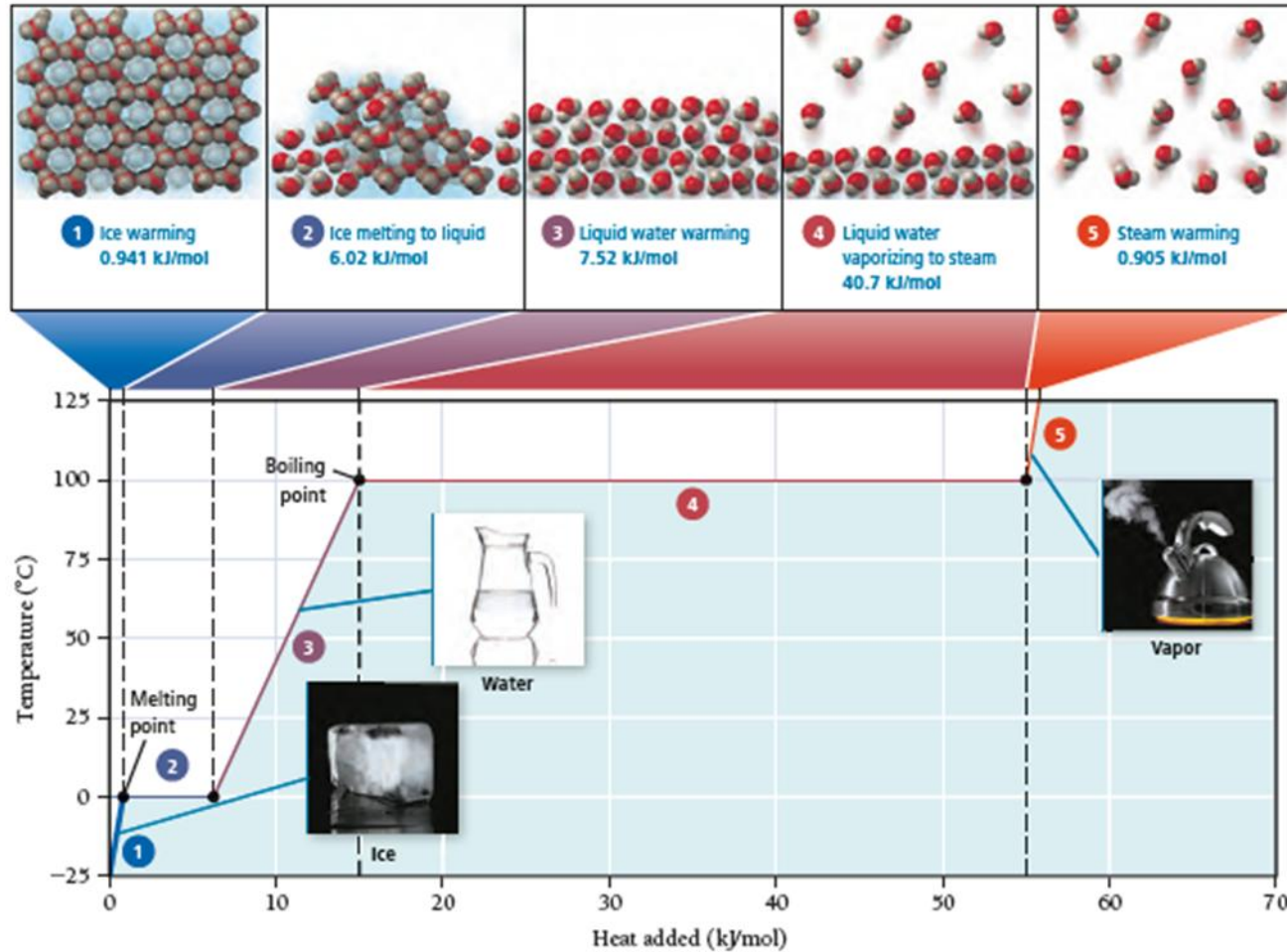


Phase Diagram

Phase Diagram for Water

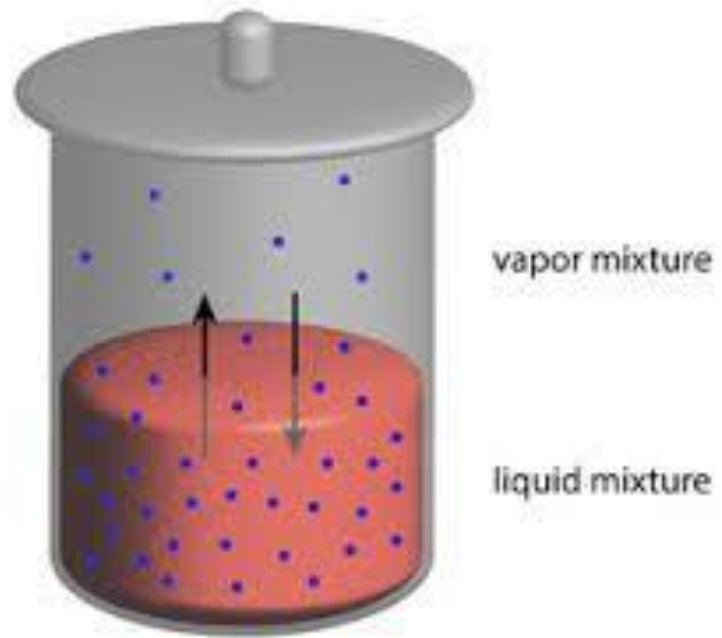


Heating Curve for Water



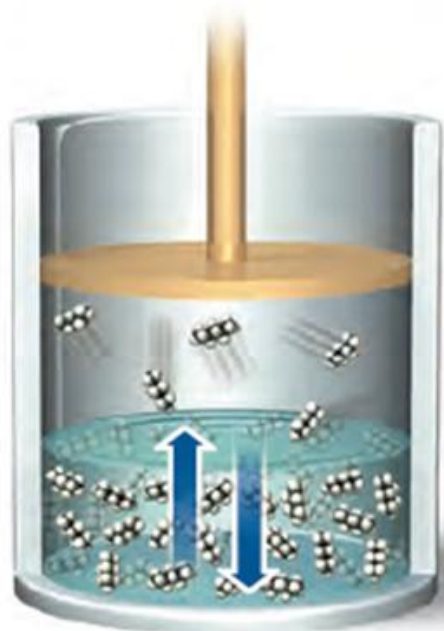
We are starting with 1 mol of ice and heating from -25°C to 125°C. The heat of fusion is 6.0 kJ/mol. The heat of vaporization is 45.3 kJ/mol. The specific heats: ice is 2.09 J/g°C, water is 4.184 J/g°C, and Steam 2.01 J/g°C.

Vapor Pressure



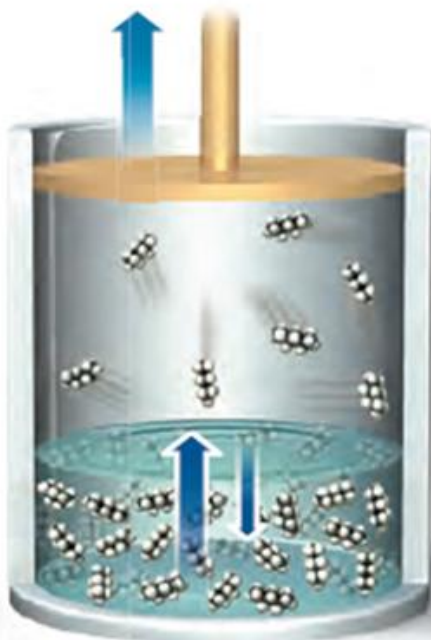
Dynamic Equilibrium

Dynamic
equilibrium



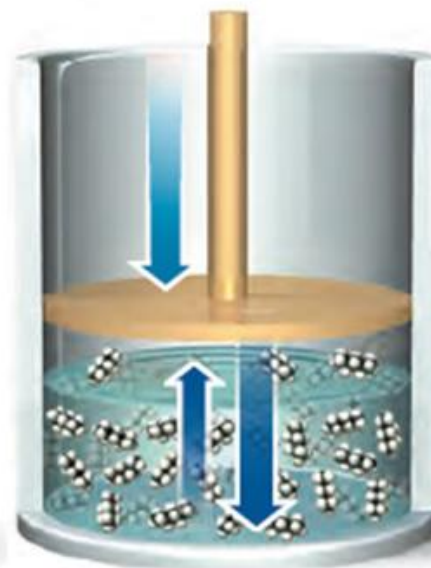
(a)

Volume is increased,
pressure falls.
More gas vaporizes,
pressure is restored.



(b)

Volume is decreased,
pressure rises.
More gas condenses,
pressure is restored.

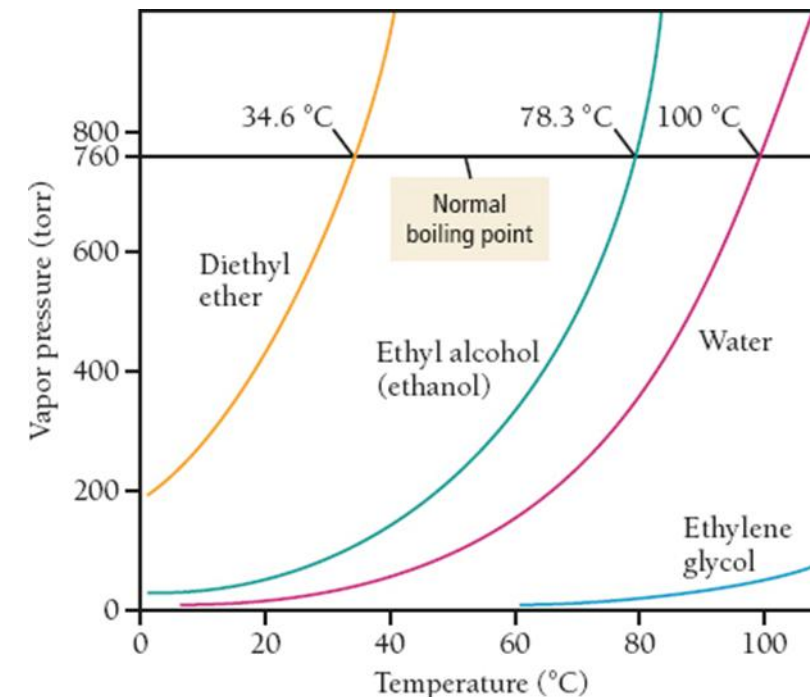


(c)

Pressure-Boiling Point

Calculate the boiling point of water at 0.7 atm.

$$\ln \frac{P_1}{P_2} = \frac{\Delta H_{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Learning Outcomes:

- 1.) Differentiate between intramolecular and intermolecular forces.
- 2.) Identify the four common types of intermolecular forces and compare their strength.
- 3.) Explain the relationship between intermolecular force strength and phase changes.
- 4.) Define vapor pressure and explain the relationships between vapor pressure and temperature.
- 5.) Define normal boiling point.
- 6.) Calculate the boiling point when the pressure is not 1 atm.
- 7.) Calculate the changes in energies when heating or cooling a pure substance.
- 8.) Identify the regions of interest on a phase diagram.
- 9.) Explain the purpose of a phase diagram.

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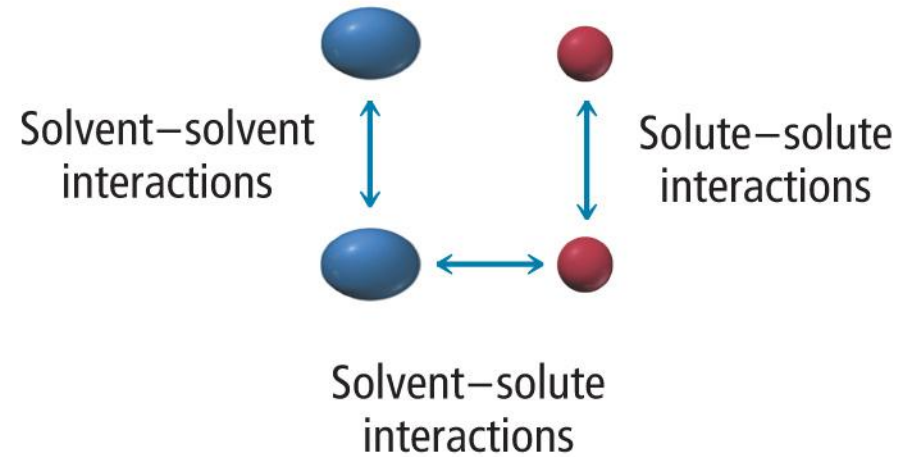
Topics:

- 1.) What are solutions?
- 2.) Discuss salt structure and calculate lattice energy.
- 3.) Explain trends in lattice energy.
- 4.) Calculate the enthalpy of dissolution for NaCl.
- 5.) Review the definition of Gibbs Free Energy and explain when Gibbs Free Energy will be negative or thermodynamically favored.
- 6.) Calculate the entropy and Gibbs Free Energy for the dissolution for NaCl.
- 7.) Discussion concentration units (molarity, ppm, molality).

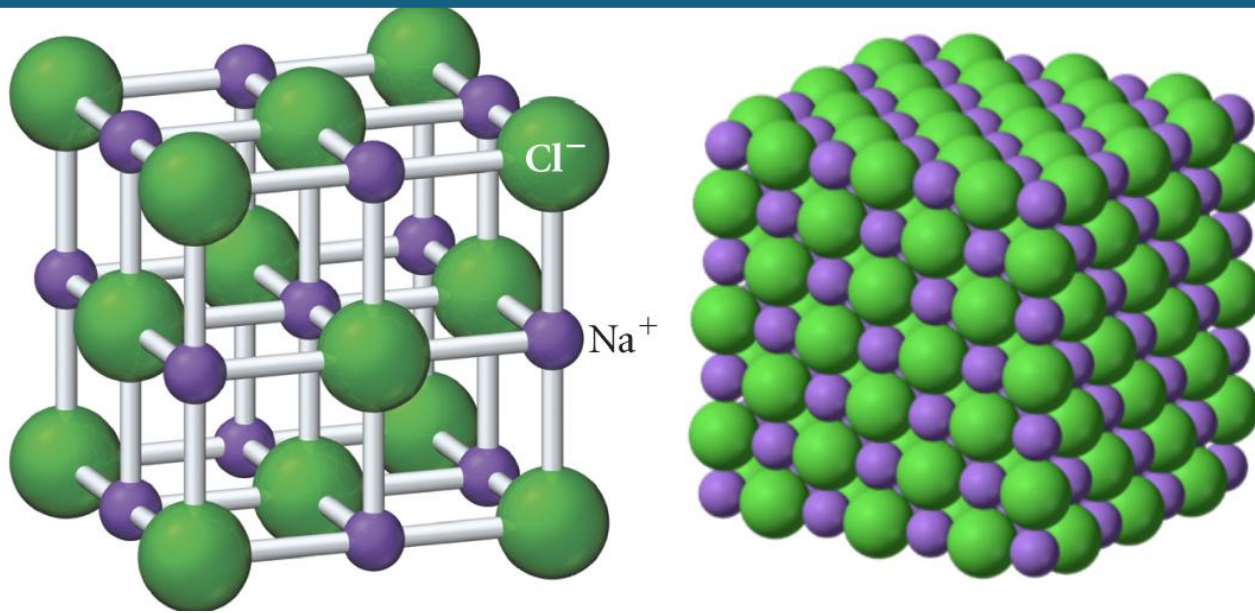


Solution Interactions

The relative strengths of these three interactions determine whether a solution forms:



Solution



Compound	Crystal Structure	ΔH (kJ/mol)
Lithium fluoride	Rock Salt	-1030
Lithium iodide	Rock Salt	-757
Potassium chloride	Rock Salt	-719
Potassium iodide	Rock Salt	-659
Magnesium fluoride	Rutile	-2922
Calcium fluoride	Fluorite	-2597

Born Haber Cycle

$$\Delta H_{\text{fusion}} (\text{Mn}) = 12.91 \text{ kJ/mol}$$

$$\Delta H_{\text{vaporization}} (\text{Mn}) = 221 \text{ kJ/mol}$$

$$\Delta H_{\text{atomization}} (\text{O}_2 \rightarrow 2 \text{ O (g)}) = 245 \text{ kJ/mol}$$

$$\Delta H_{\text{first EA}} (\text{O}) = -142 \text{ kJ/mol}$$

$$\Delta H_{\text{second EA}} (\text{O}) = 844 \text{ kJ/mol}$$

$$\text{Heat of Formation: } -520 \text{ kJ/mol}$$

$$\text{Mn (1}^{\text{st}} \text{ IE): } 717.3 \text{ kJ/mol}$$

$$\text{Mn (2}^{\text{nd}} \text{ IE): } 1509.0 \text{ kJ/mol}$$



Born Haber Cycle

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$\text{Mn (s)} \rightarrow \text{Mn (l)}$

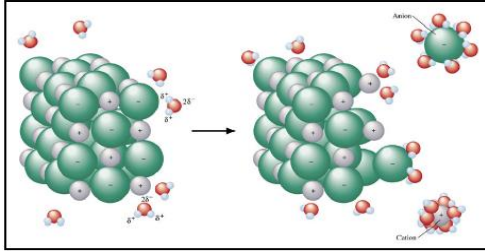
$\text{Mn (l)} \rightarrow \text{Mn (g)}$

$\text{O (g)} \rightarrow \text{O}^- (\text{g})$

$\text{O}^- (\text{g}) \rightarrow \text{O}^{2-} (\text{g})$

What “attractions” are broken?

What “attractions” are created?



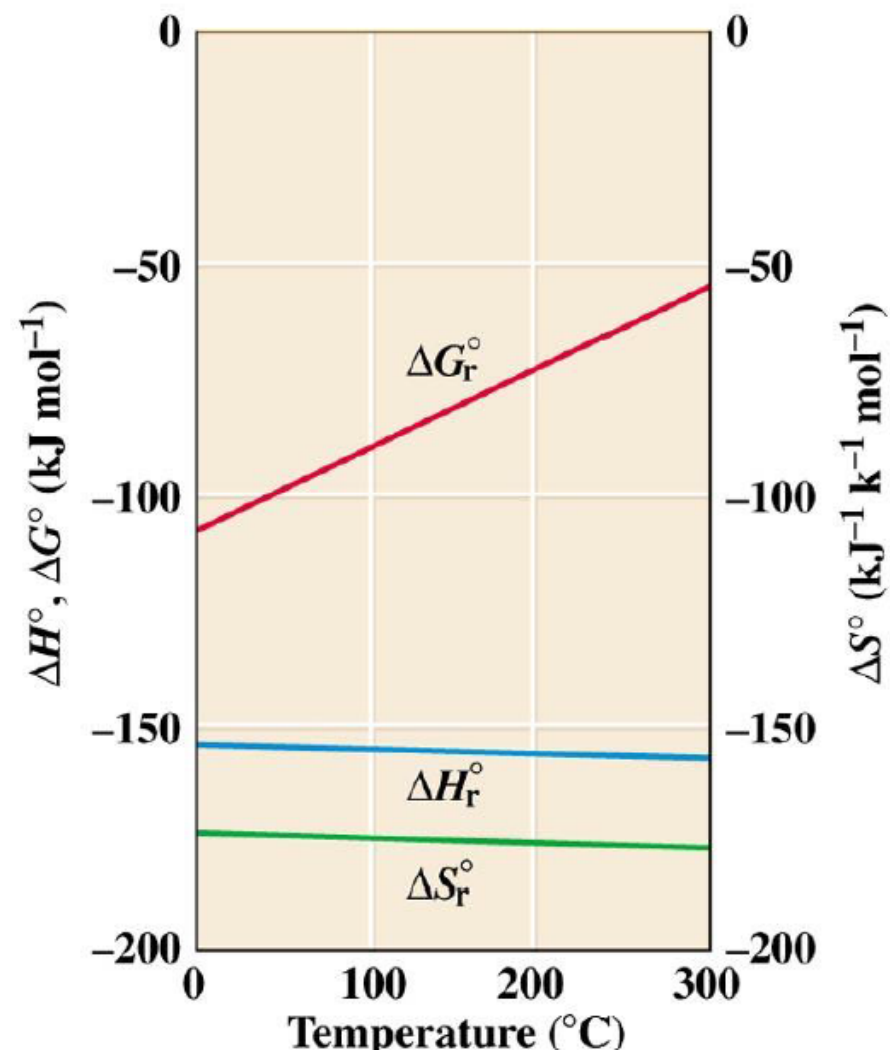
What is the driving force?

How can we rationalize from a
thermodynamic standpoint?

Lattice Energy for Sodium Chloride: **-772 kJ/mol NaCl**

What is the enthalpy of dissolution?



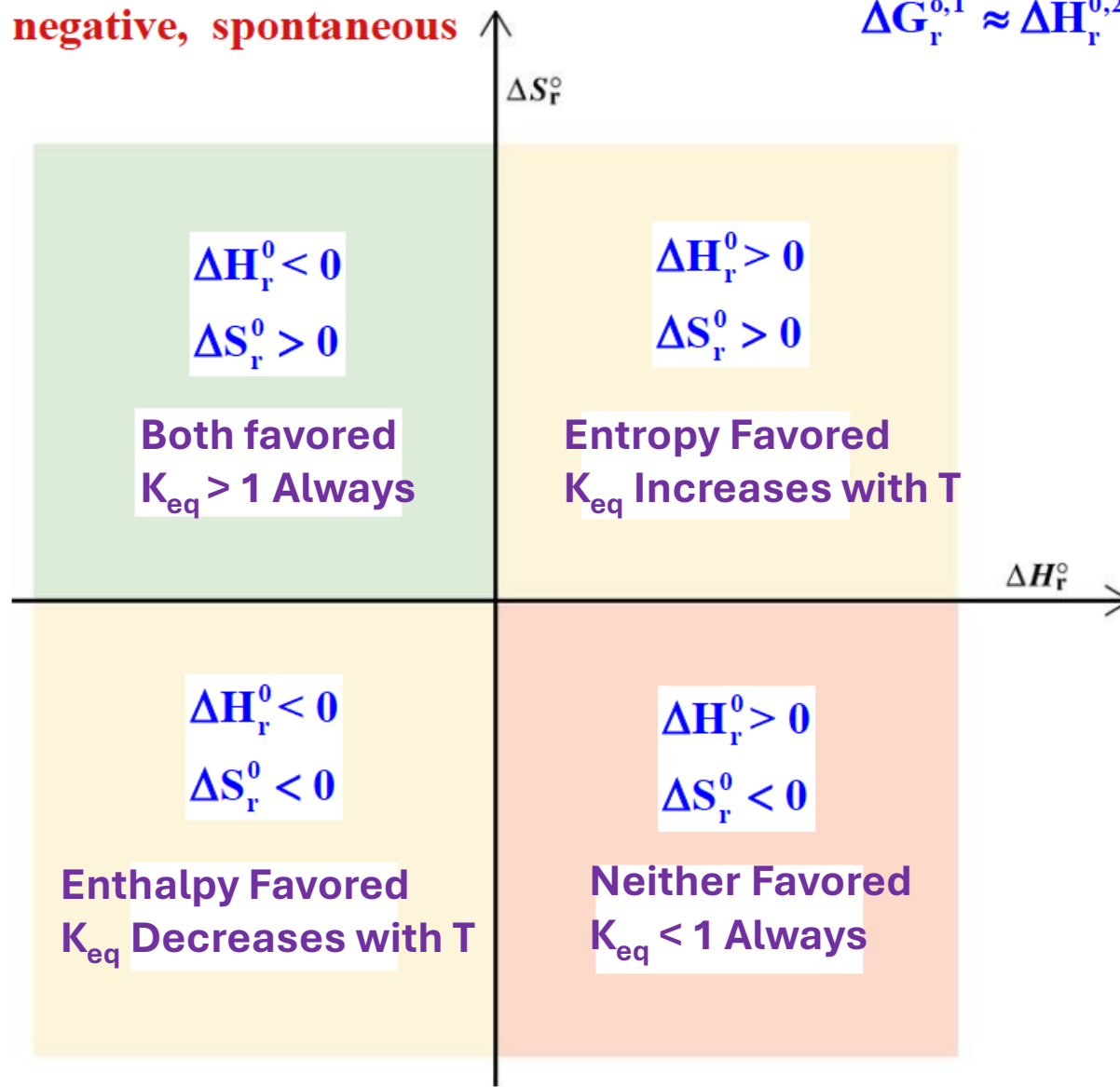


$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K_{\text{eq}}$$

Spontaneous Reaction - Influence of Temperature

$\Delta G_r < 0 \Rightarrow$ negative, spontaneous

$$\Delta G_r^{0,T} \approx \Delta H_r^{0,25C} - T\Delta S_r^{0,25C}$$



Dissolution Calculation

NaCl (s): $S^\circ = 72.1 \text{ J/mol NaCl K}$

Na⁺ (aq): $S^\circ = 58.5 \text{ J/mol Na}^+ \text{ K}$

Cl⁻ (aq): $S^\circ = 56.5 \text{ J/mol Cl}^- \text{ K}$

Part Per Million (ppm)

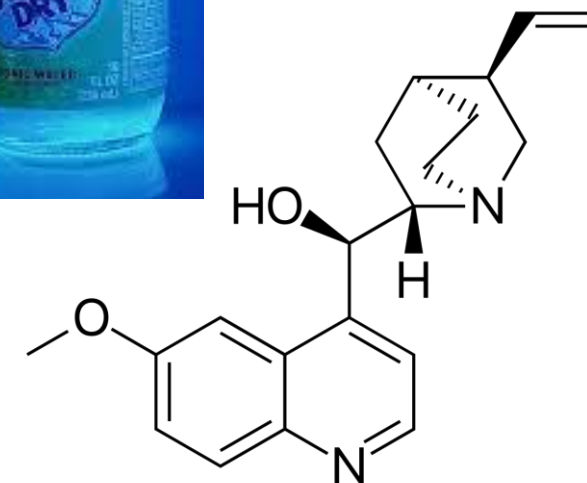
The quinine concentration in tonic water is 83 ppm.

What is the molarity of quinine in tonic water?

The total volume is 100-mL. Molar Mass = 324.42 g/mol



Reddit

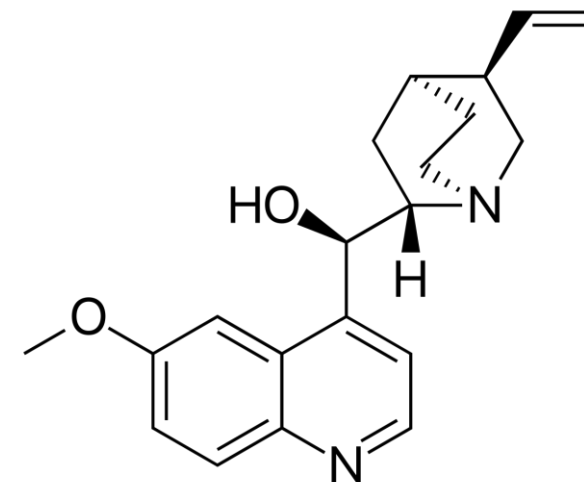


More Concentration Units!!!

From the molarity of quinine in tonic water, what is the molality?



Reddit



Learning Outcomes:

- 1.) Define solute, solvent, and solution.
- 2.) Define lattice energy and identify factors that influence lattice energy (or the strength of the ionic bond).
- 3.) Create a Born-Haber cycle to compute lattice energy.
- 4.) Compute ΔH , ΔS , and ΔG for dissolution.
- 5.) Calculate molarity, molality, and ppm for solutions.

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Topics:

- 1.) Colligative Properties
- 2.) Raoult's Law
- 3.) Dalton's Law of Gaseous Mixtures
- 4.) Boiling Point Elevation and Freezing Point Depression
- 5.) Osmotic Pressure

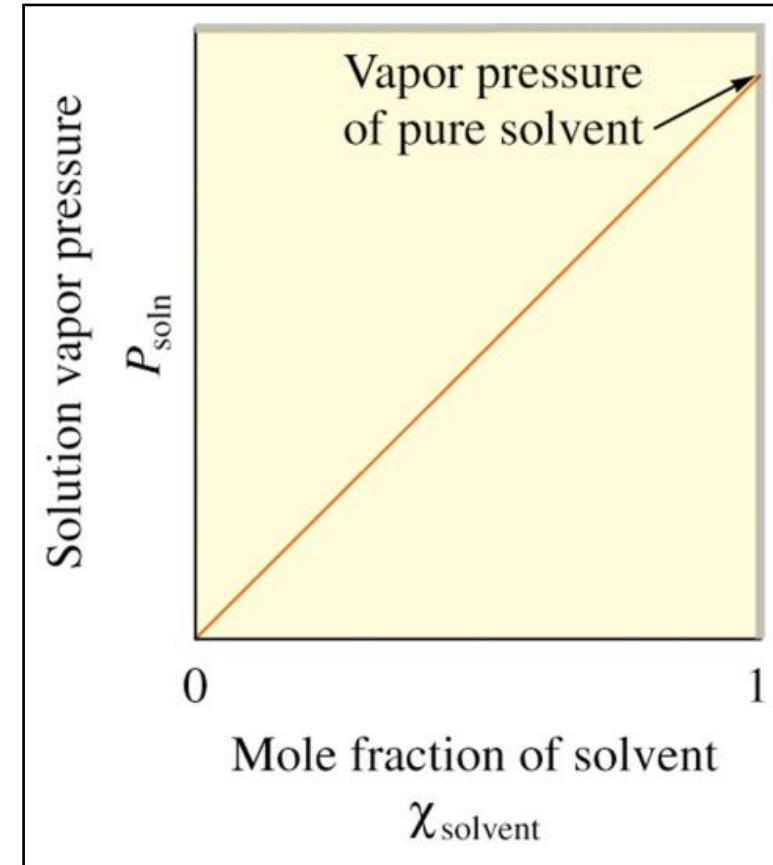
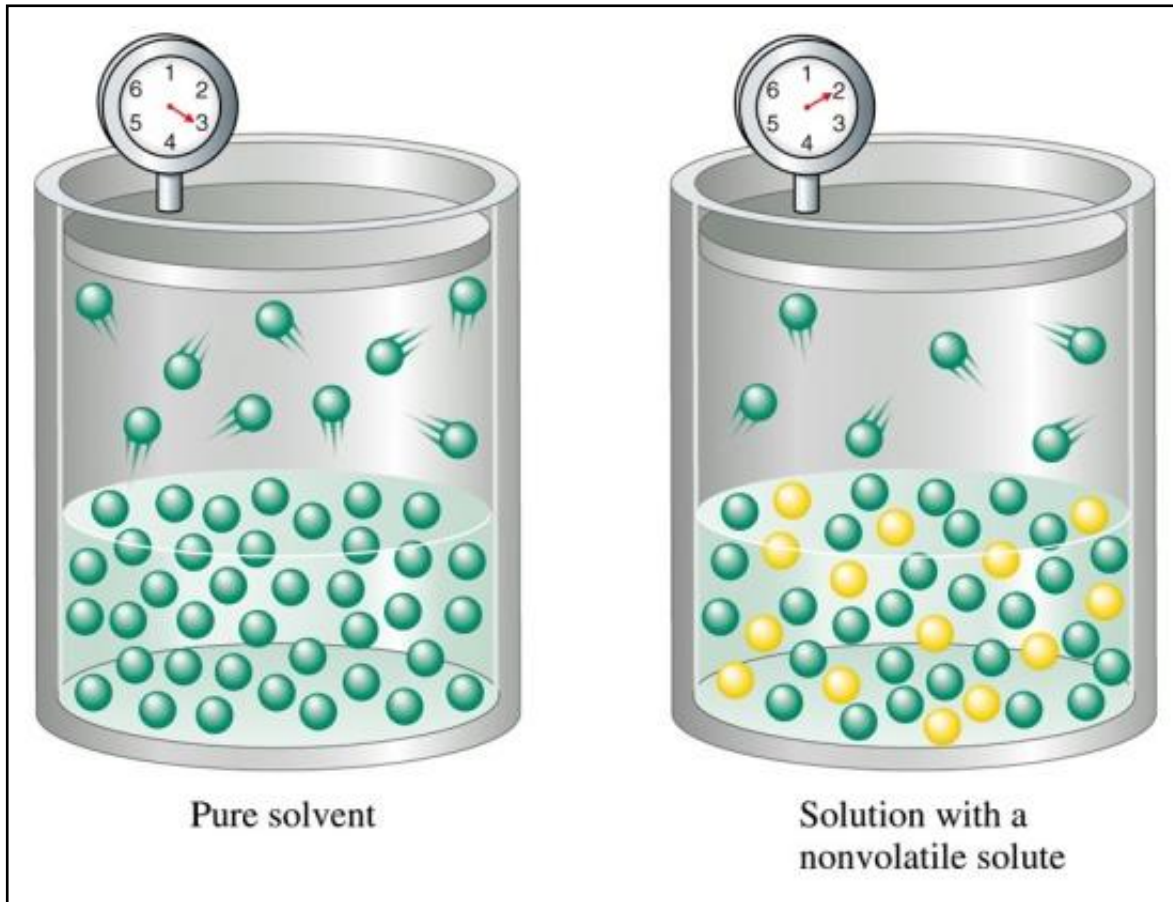


Colligative Properties



Solution

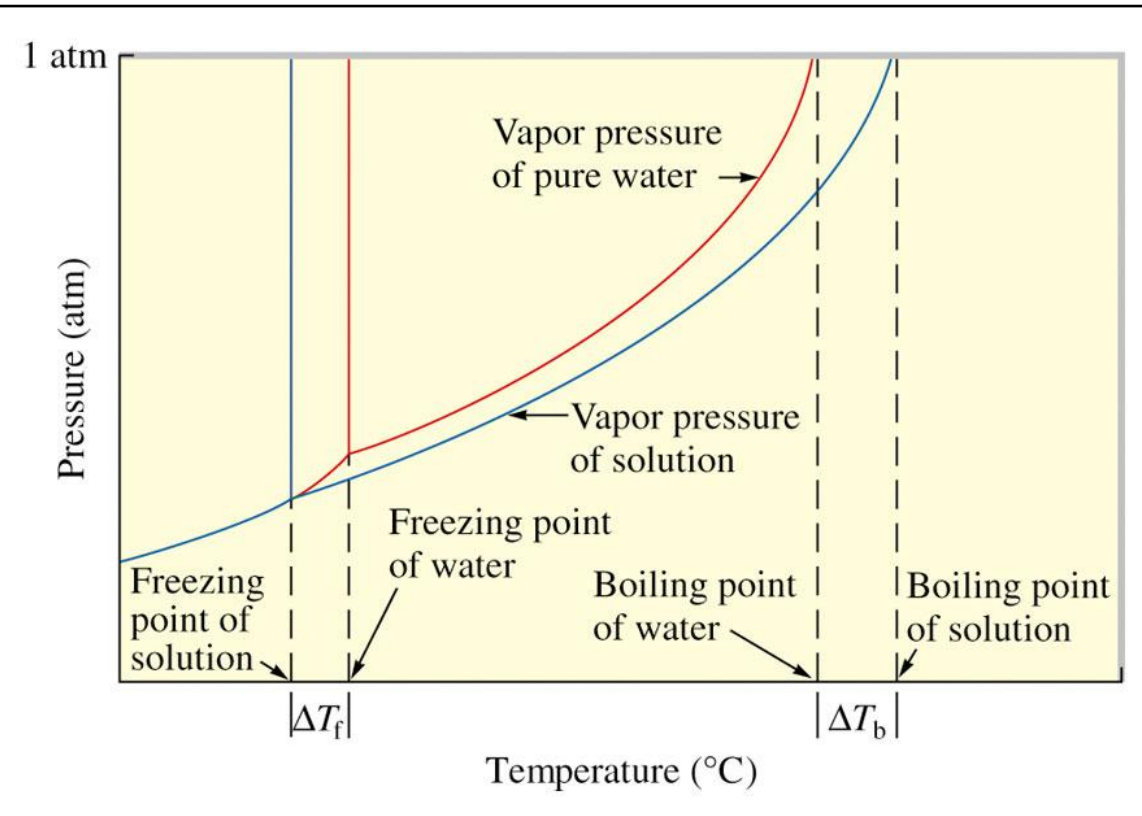
Raoult's Law



Consider a mixture containing two chlorofluorocarbons (CFCs) that have been synthesized in the laboratory. The molecules have been synthesized in the flask as shown below. There are no remaining starting materials. The two molecules, C_2FCl_5 and $\text{C}_2\text{F}_2\text{Cl}_4$, have vapor pressures of 20.00 mm Hg and 140.0 mm Hg at 25.0 °C, respectively. The two molecules are each present in a 1:1 mole ratio in solution. Calculate the vapor pressure of each and the composition in the vapor phase. These are the only two components contributing to the vapor pressure. The solution is ideal.

Recall Dalton's Law: $P_{\text{total}} = P_1 + P_2 + \dots + P_n$

Molal Boiling-Point Elevation and Freezing-Point Depression Constants for Select Solvents



$$K_f = \frac{RT_f^2(MW)}{\Delta H_{fus}}$$

$$K_b = \frac{RT_b^2(MW)}{\Delta H_{vap}}$$

Solvent	Boiling Point (°C)	K_b (°C kg/mol)	Freezing Point (°C)	K_f (°C kg/mol)
Water (H ₂ O)	100.0	0.51	0.	1.86
Carbon tetrachloride (CCl ₄)	76.5	5.03	−22.99	30.
Chloroform (CHCl ₃)	61.2	3.63	−63.5	4.70
Benzene (C ₆ H ₆)	80.1	2.53	5.5	5.12
Carbon disulfide (CS ₂)	46.2	2.34	−111.5	3.83
Ethyl ether (C ₄ H ₁₀ O)	34.5	2.02	−116.2	1.79
Camphor (C ₁₀ H ₁₆ O)	208.0	5.95	179.8	40.

Example

Calculate the boiling point of a solution of 400.0 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) dissolved in 400.0 g of water. $K_f = 1.86^\circ\text{C}/\text{m}$ and $K_b = 0.512^\circ\text{C}/\text{m}$. Use 100°C as the boiling point of water.

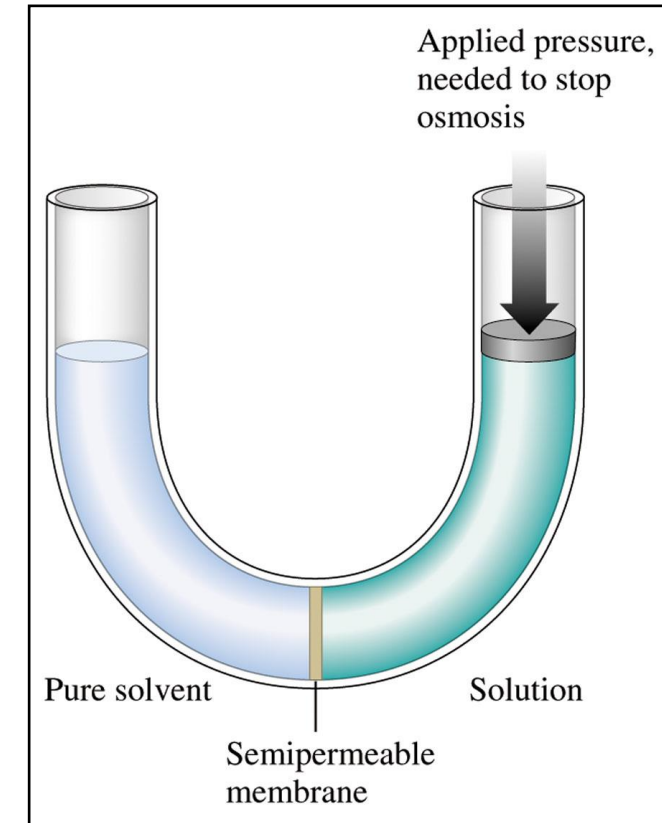
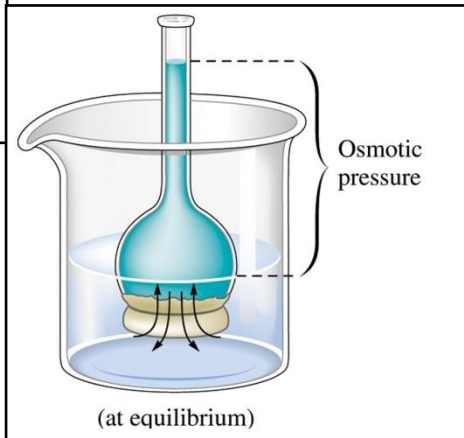
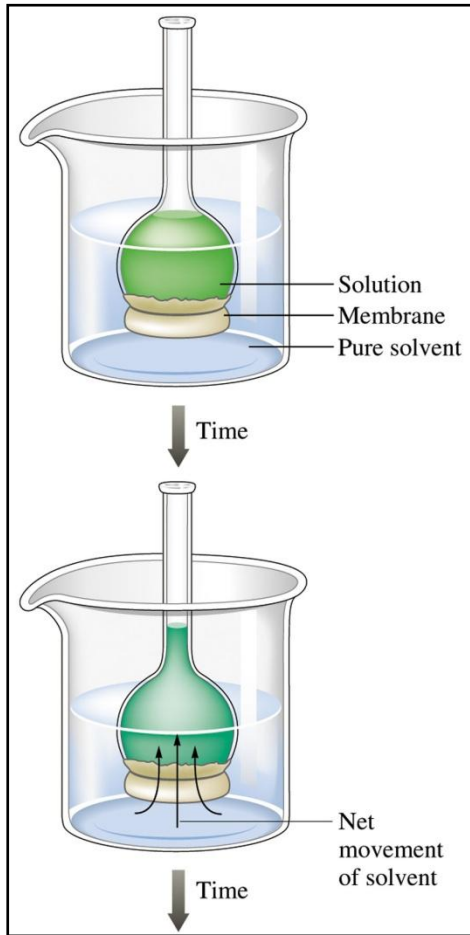
TABLE 17.6 Expected and Observed Values of the van't Hoff Factor for 0.05 *m* Solutions of Several Electrolytes

Electrolyte	<i>i</i> (expected)	<i>i</i> (observed)
NaCl	2.0	1.9
MgCl ₂	3.0	2.7
MgSO ₄	2.0	1.3
FeCl ₃	4.0	3.4
HCl	2.0	1.9
Glucose*	1.0	1.0

*A nonelectrolyte shown for comparison.

Calculate boiling point of an aqueous 1.83 *m* (NH₄)₂SO₄ (molar mass = 132.15 g/mol) solution.

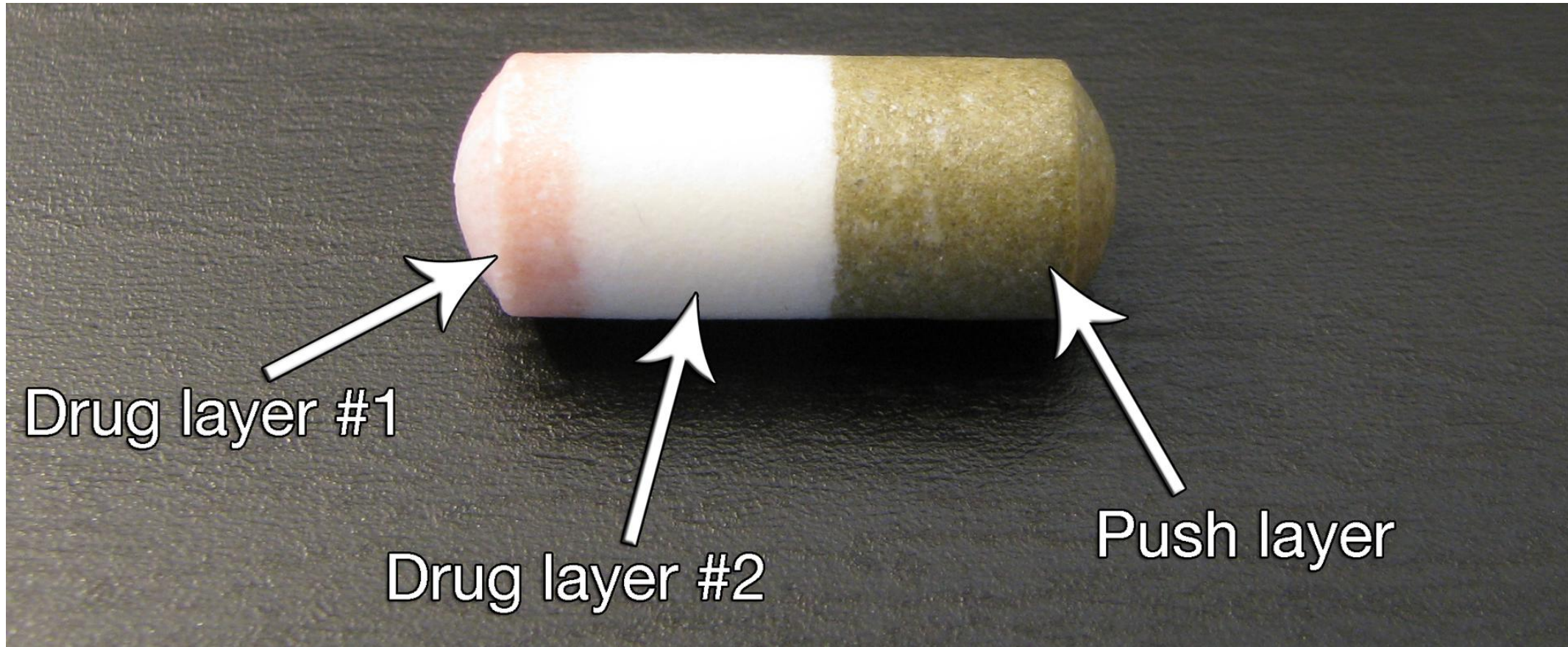
Osmotic Pressure



Osmotic Pressure Example

A nonelectrolyte compound is found to have a molar mass of 598 g/mol. If 33.7 mg of the compound is dissolved in enough water to make 175 mL of solution at 25°C, what is the osmotic pressure of the resulting solution?

Applications of Osmosis



CHEM 210

Modern Applications of Chemistry

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Topics:

- 1.) Solubility Equilibrium: Reactions and K_{sp}
- 2.) Reaction Quotients (Q), Equilibrium, and Thermodynamics
- 3.) Properties of Equilibrium Constants
- 4.) Calculations with K_{sp}
- 5.) Define molar solubility
- 6.) The Common Ion Effect

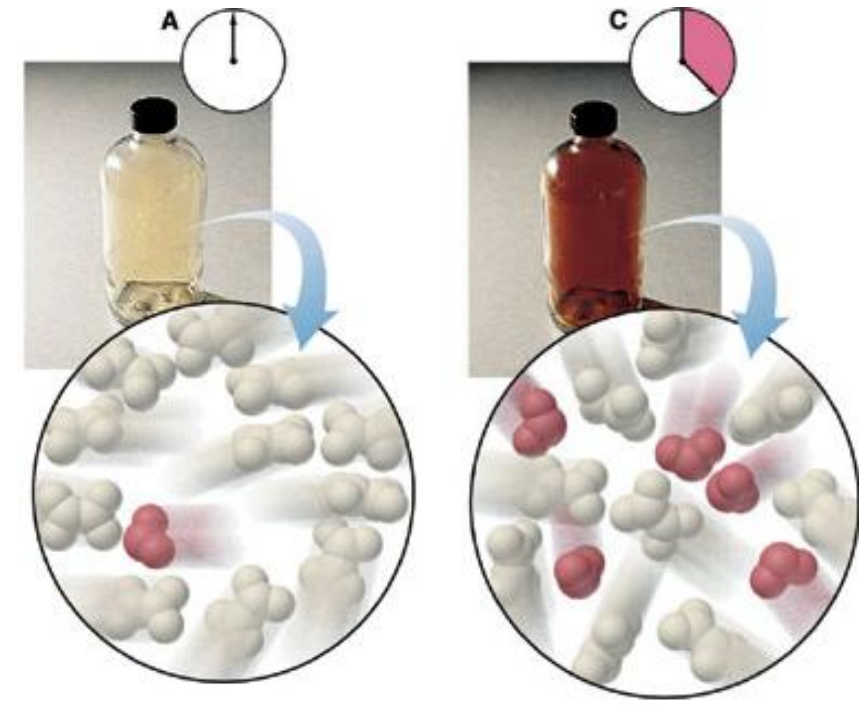
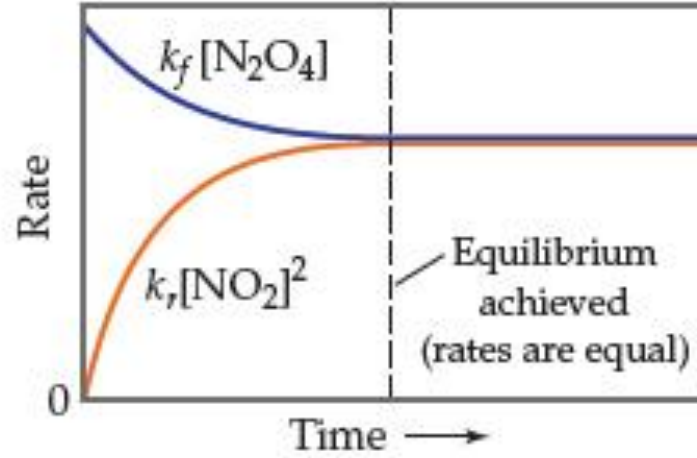
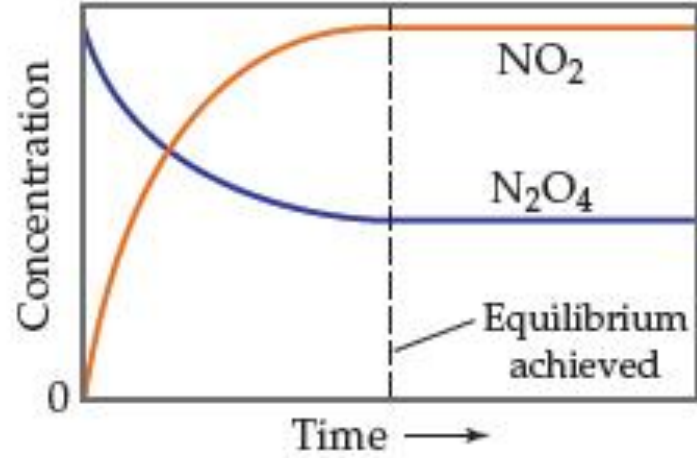


Solubility Equilibrium

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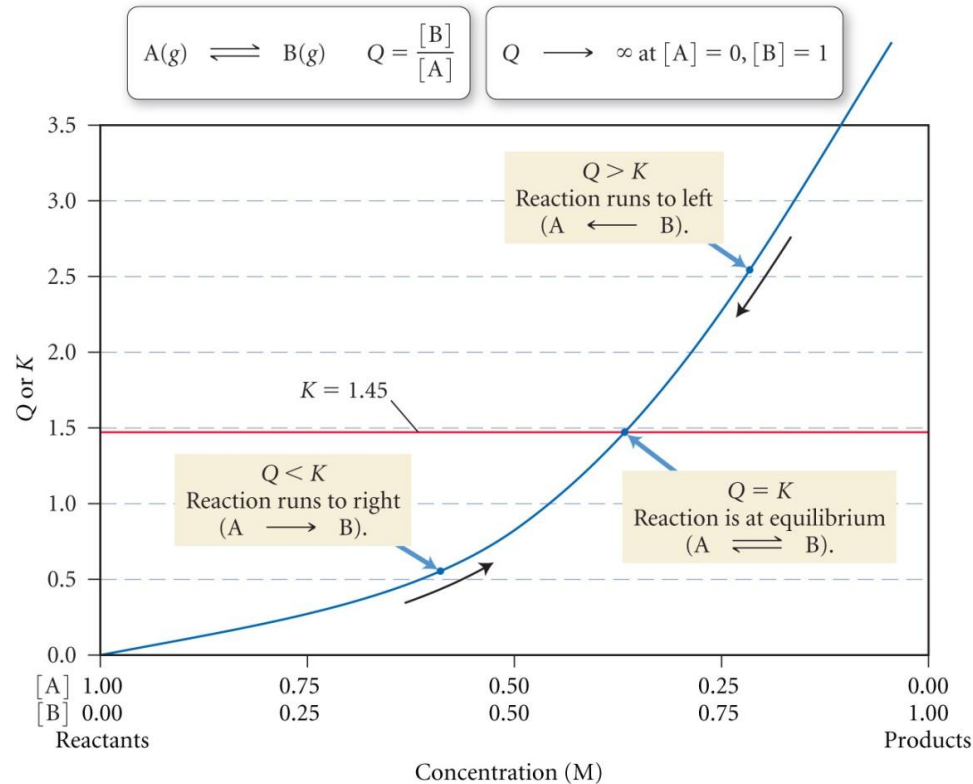
Chemical Equilibrium



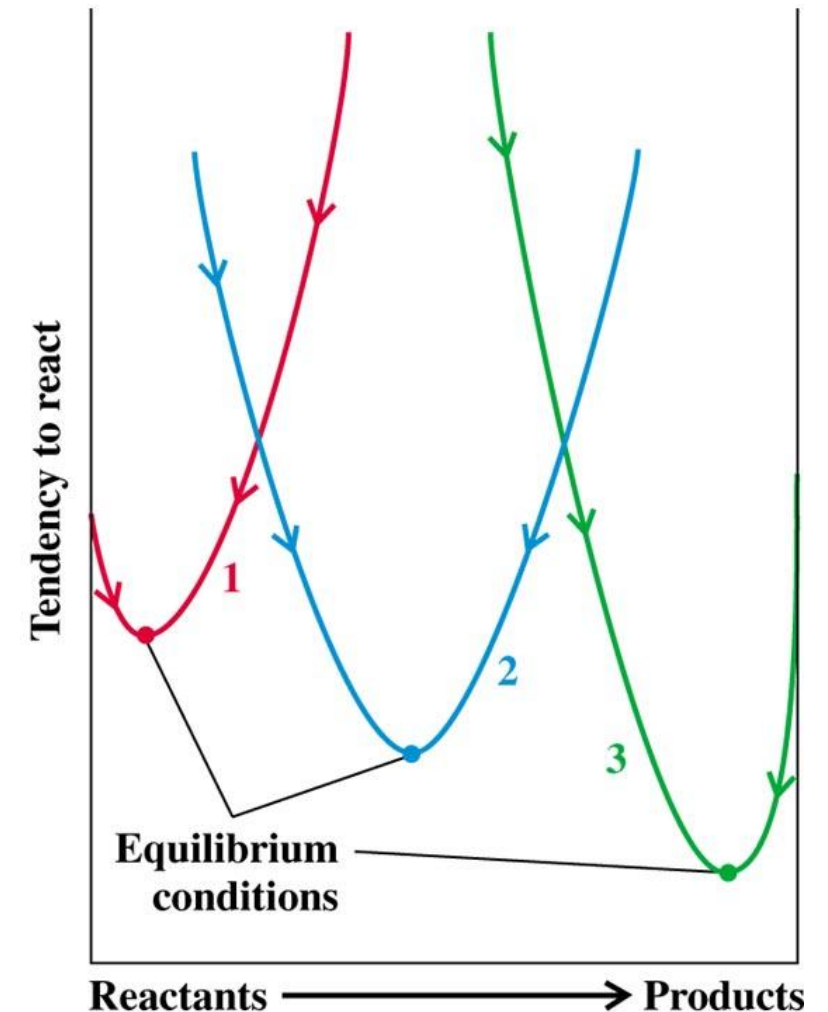
$$\Delta G^{\circ} = - RT \ln K$$

The Direction of a Reaction

Q , K , and the Direction of a Reaction



$$\Delta G = \Delta G^\circ + RT \ln Q$$



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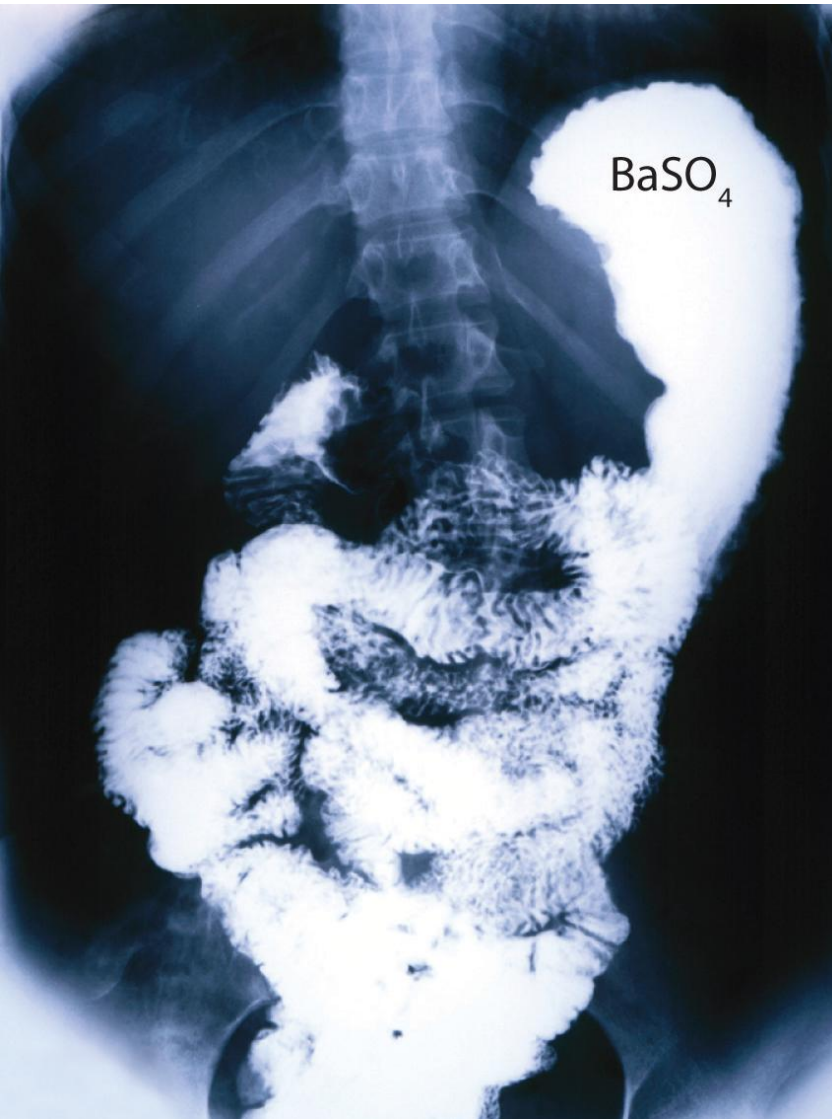
$$\Delta G^\circ = -RT \ln K$$

1.] Forward vs. Reverse Reactions

2.] A Change in the Coefficients

3.] Multistep Reactions

Barium Sulfate Equilibrium



Addition of Barium Sulfate in Water:

Equilibrium Reaction that Forms:

Solubility Product Constant, K_{sp}

Barium sulfate is opaque to X-rays.

Solubility Problem



Determine the **mass of lead(II) iodate** (MW = 557 g/mol) dissolved in 2.50 L of a saturated aqueous solution of $\text{Pb}(\text{IO}_3)_2$ at 25°C. The K_{sp} of $\text{Pb}(\text{IO}_3)_2$ is 2.6×10^{-13} .

Common Ion Example

Given, Thallium(I) iodate ($K_{sp} = 3.1 \times 10^{-6}$ at 25°C), determine the molar solubility in $0.050 \text{ mol L}^{-1} \text{ KIO}_3$ at 25°C .

Precipitation

A solution is prepared by adding **0.23 g** of PbCl_2 (**$K_{\text{sp}} = 1.6 \times 10^{-5}$**) to a 100-mL aqueous solution. Will a precipitate form?